
PRASAD'S PHYSICS**BRAHMAPUR****11th - (Assignment)**

Scalars and Vectors**Numericals**

01. The angle between two vectors of equal magnitude is 120° . Prove that the magnitude of their resultant is equal to either of them.

02. The resultant of two equal forces acting at right angles to each other is 1414 N. Find the magnitude of each force.

Ans. $P = Q = 1000 \text{ N}$

03. What should be the angle between two vectors if the magnitude of their resultant is equal to the difference between their magnitudes.

Ans. $\theta = 180^\circ$

04. A vector $\vec{A}$ makes an angle of 20° and $\vec{B}$ makes an angle of 110° with the X-axis. The magnitudes of these vectors are 3m and 4m respectively. Find the resultant.

Ans. 5m

05. Two vectors acting at 90° to each other are increased twice each, find the ratio of new resultant to previous one.

Ans. 2 : 1

06. Two forces acting in opposite directions have a resultant of 10N. If they act at right angles to each other, the resultant is 50 N. Find magnitudes of the forces.

Ans. $B = 30 \text{ N}$

07. Given that $\vec{A} + \vec{B} + \vec{C} = 0$. Out of three vectors two are equal in magnitude and the magnitude of the third vector is $\sqrt{2}$ times that of either of the two. Find the angle between vectors.

Ans. $\theta = 90^\circ$

08. If the magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively and if $\vec{A} + \vec{B} = \vec{C}$, find the angle between $\vec{A}$, and $\vec{C}$.

Ans. $\theta = \cos^{-1}(3/5)$

09. The magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 12, 5 and 13 units respectively and $\vec{A} + \vec{B} = \vec{C}$, find the angle between $\vec{A}$ and $\vec{B}$.

Ans. $\vec{A}$ and $\vec{B}$ is 90°

10. Add vector $\vec{A}$, $\vec{B}$ and $\vec{C}$ each having magnitude of 100 unit and inclined to the X-axis at angles 45° , 135° and 315° respectively.

Ans. 100

11. A force 100 N is resolved in to two equal components at 60° to each others find the components.

Ans. $x = \frac{100}{\sqrt{3}}$

12. Find the unit vector of $\vec{A} = \hat{i} + \hat{j} + \hat{k}$.

Ans. $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = (\hat{i} + \hat{j} + \hat{k})/\sqrt{3}$

13. What is angle made by vector $\vec{A} = 2\hat{i} + 2\hat{j}$ with x-axis?

Ans. $\theta = 45^\circ$

14. Determine the vector which when added to the resultant of $\vec{A} = 3\hat{i} - 5\hat{j} + 7\hat{k}$ and $\vec{B} = 2\hat{i} + 4\hat{j} - 3\hat{k}$ gives unit vector along y-direction.

Ans. $-5\hat{i} + 2\hat{j} - 4\hat{k}$.

15. Find the angle between the vectors

$\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 5\hat{i} - 2\hat{j} + 3\hat{k}$.

Ans. 69.5° .

16. If the magnitude of two vectors are 3 and 4 and the magnitude of their scalar product is 6, find the angle between the vectors.

Ans. $\theta = 60^\circ$

17. Prove that vectors $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j}$ are perpendicular to each other.

18. A particle moves from position $(3\hat{i} + 2\hat{j} - 6\hat{k})$ to a position $(14\hat{i} + 13\hat{j} + 9\hat{k})$ in metre units and a constant force $(4\hat{i} + 2\hat{j} + 3\hat{k})$ Newtons acts on it. Calculate the work done by the force.

Ans. 111J

19. Find $\vec{A} \times \vec{B}$ if $\vec{A} = \hat{i} - \hat{j}$ and $\vec{B} = \hat{j} - \hat{i}$.

Ans. zero

20. The angle between vectors $\vec{A}$ and $\vec{B}$ is 60° . What is the ratio $\vec{A} \cdot \vec{B}$ and $|\vec{A} \times \vec{B}|$?

Ans. $\frac{1}{\sqrt{3}}$

Multiple Choice Questions

01. The resultant of $\vec{A}$ and $\vec{B}$ makes an angle α with $\vec{A}$ and β with $\vec{B}$:

- (a) $\alpha < \beta$ (b) $\alpha < \beta$ if $A < B$
(c) $\alpha < \beta$ if $A > B$ (d) $\alpha < \beta$ if $A = B$

Ans. (c)

02. A vector is not changed if :

- (a) It is rotated through an arbitrary angle
(b) It is multiplied by an arbitrary scalar
(c) It is across multiplied
(d) It is slid parallel to itself

Ans. (d)

03. A vector $\vec{A}$ points vertically upwards and $\vec{B}$ points towards north. The vector product $\vec{A} \times \vec{B}$ is :

- (a) Along west (b) Along east
(c) Zero (d) Vertically downward

Ans. (a)

04. Let $\vec{C} = \vec{A} + \vec{B}$ is :

- (a) $|\vec{C}|$ is always greater than $|\vec{A}|$
(b) It is possible to have $|\vec{C}| < |\vec{A}|$ & $|\vec{C}| < |\vec{B}|$
(c) C is always equal to $A + B$
(d) C is never equal to $A + B$

Ans. (b)

05. If vectors $\vec{a} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} - x\hat{k}$ are to be perpendicular to each other, the value of x should be :

- (a) 2 (b) -2 (c) 3 (d) -3

Ans. (a)

06. A particle moves from the point (1, 0, 2.5) to the point (-2, 3, 4) m when a force $\vec{F} = (\hat{i} + 4\hat{k})\text{N}$ acts on it. The work done on it is :

- (a) 6J (b) 30 J (c) 3J (d) 9J

Ans. (c)

07. If the vector sum of two unit vectors is a unit vector, then the magnitude of the difference is :

- (a) 1 (b) $\sqrt{2}$ (c) $1/\sqrt{2}$ (d) $\sqrt{3}$

Ans. (d)

08. Moment of inertia is a :

- (a) scalar (b) vector
(c) scalar and vector (d) tensor

Ans. (d)

09. The maximum value of magnitude of $(\vec{A} - \vec{B})$ is :

- (a) $A - B$ (b) A
(c) $A + B$ (d) $\sqrt{A^2 + B^2}$

Ans. (c)

10. The resultant of two forces 5 Newton and 10 Newton can never be :

- (a) 12 N (b) 5 N (c) 10 N (d) 4 N

Ans. (d)

11. If two non-zero vectors $\vec{A}$ and $\vec{B}$ obey the relation

$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|, \text{ the angle between } \vec{A} \text{ and } \vec{B} \text{ is :}$$

- (a) 0° (b) 45° (c) 90° (d) 180°

Ans. (c)

12. Angular momentum is :

- (a) scalar (b) axial vector
(c) scalar and vector only
(d) tensor

Ans. (b)

13. Given vector $\vec{R} = 2\hat{i} + 3\hat{j}$. The angle between $\vec{R}$ and Y-axis is :

- (a) $\tan^{-1}(3/2)$ (b) $\tan^{-1}(2/3)$
(c) $\sin^{-1}(2/3)$ (d) $\cos^{-1}(2/3)$

Ans. (b)

14. The magnitude of vector $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively. If $\vec{A} + \vec{B} = \vec{C}$, the angle between $\vec{A}$ and $\vec{C}$ is :

- (a) $\pi/2$ (b) $\cos^{-1}(0.6)$
(c) $\tan^{-1}(0.75)$ (d) $\pi/4$

Ans. (b)

15. If the vector sum of two unit vectors is a unit vector, then the magnitude of the difference is :

- (a) 1 (b) $\sqrt{2}$ (c) $1/\sqrt{2}$ (d) $\sqrt{3}$

Ans. (d)

16. Two forces with equal magnitude $\vec{F}$ act on a body and the magnitude of the resultant force $F/3$. The angle between the two forces is :

- (a) $\cos^{-1}(-17/18)$ (b) $\cos^{-1}(-1/3)$
(c) $\cos^{-1}(2/3)$ (d) $\cos^{-1}(8/9)$

Ans. (a)

17. The radius vector and linear momentum of a particle are given by $\vec{r} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{p} = 2\hat{i} - 3\hat{j} + \hat{k}$. The angular momentum of particle about origin is :

- (a) $2\hat{i} - 4\hat{j}$ (b) $4\hat{i} - 8\hat{k}$
(c) $2\hat{i} - 4\hat{j} + 2\hat{k}$ (d) $4\hat{i} - 8\hat{j}$

Ans. (b)

18. The angle made by vector $\vec{A}$ and $\vec{B}$ is θ . The value of the triple product $\vec{A} \cdot (\vec{B} \times \vec{A})$ is :

- (a) A^2B (b) $A^2B \sin \theta$
(c) $A^2B \cos \theta$ (d) 0

Ans. (d)

19. A force of 120 N and a force of 20N acting simultaneously at a point may produce a resultant force of :

- (a) 80N (b) 140N (c) 160N (d) None

Ans. (b)

20. The minimum number of unequal forces whose vector sum can be zero is :

- (a) 2 (b) 5 (c) 3 (d) 4

Ans. (c)

21. The vector sum of two forces is perpendicular to their vector difference. In this case the forces are :

- (a) Equal to each other
(b) Equal to each other in magnitude
(c) Not equal to each other in magnitude
(d) Can not be predicted

Ans. (b)

22. Two forces of magnitude P and $\sqrt{3}P$ act at right angles to each. Their resultant makes an angle θ with P. Which of the following gives the correct value of θ .

- (a) 30° (b) 45° (c) 60° (d) None

Ans. (c)

23. If $\vec{a}_1$ and $\vec{a}_2$ are non – collinear unit vectors and if $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then the value of $(2\vec{a}_1 - 5\vec{a}_2) \cdot (3\vec{a}_1 + \vec{a}_2)$ is :

- (a) 41/2 (b) 11/2 (c) $-(11/2)$ (d) $-(41/1)$

Ans. (c)

24. The express $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is a :

- (a) Unit vector (b) Null vector
(c) Vector of magnitude $\sqrt{2}$ (d) scalar

Ans. (a)

25. The numerical value of $\vec{A} = 4\hat{i} - 3\hat{j} + 5\hat{k}$ is :

- (a) $5\sqrt{2}$ (b) 2 (c) 6 (d) $\sqrt{3}$

Ans. (a)

26. The angle between two vectors $-2\hat{i} + 3\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$ is :

- (a) 0 (b) 45° (c) 90° (d) 180°

Ans. (c)

27. Three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ satisfy the relation, $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \cdot \vec{C} = 0$, if $\vec{B}$ and $\vec{C}$ are not lying in the same plane, then $\vec{A}$ is parallel to :

- (a) $\vec{B}$ (b) $\vec{C}$ (c) $\vec{B} \times \vec{C}$ (d) $\vec{B} \cdot \vec{C}$

Ans. (c)

28. Consider a vector $\vec{F} = 4\hat{i} - 3\hat{j}$. Another vector that is perpendicular to F is :

- (a) $4\hat{i} + 3\hat{j}$ (b) $6\hat{i}$ (c) $7\hat{k}$ (d) $3\hat{i} - 4\hat{j}$

Ans. (c)

29. The maximum and minimum magnitude of the resultant of two vectors are 17 units and 7 units respectively. If these two vectors are at right angles to each other, the magnitude of their resultant is :

- (a) 14 (b) 16 (c) 13 (d) 18

Ans. (c)

30. Out of the following sets of forces the resultant of which can not be 0.

- (a) 10,10,10 (b) 10,10,20
(c) 10,20,20 (d) 10,20,40

Ans. (d)

31. Give that $\vec{A} + \vec{B} = \vec{A} - \vec{B}$. This can be true when :

- (a) $\vec{A} = 0$ (b) $\vec{A}$ is perpendicular to $\vec{B}$
(c) $\vec{B} = 0$ (d) Neither $\vec{A}$ nor $\vec{B}$ is a null vector.

Ans. (c)

32. What is the component to a force of 10N in a direction perpendicular to it.

- (a) 10 N (b) Zero
(c) $10 \times 1/\sqrt{2}$ N (d) $10 \times \sqrt{2}$ N

Ans. (b)

33. If $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$ then angle between A and B is :

- (a) 45° (b) 30° (c) 60° (d) 90°

Ans. (a)

34. If $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$ then magnitude and direction of $\vec{A} + \vec{B}$ with x-axis will be :

- (a) 5, $\tan^{-1}(3/4)$ (b) $5\sqrt{5}$, $\tan^{-1}(1/2)$
(c) $25 \tan^{-1}(3/4)$ (d) 10, $\tan^{-1}(5)$

Ans. (b)

35. If two vectors $2\hat{i} + 3\hat{j} + \hat{k}$ and $-4\hat{i} - 6\hat{j} - \pi\hat{k}$ are parallel to each other than value of π is :

- (a) 0 (b) 2 (c) 3 (d) 4

Ans. (b)

36. A man walks 20m at an angle 60° north of east. How far towards east has he traveled :

- (a) 10 m (b) $20\sqrt{3}$ m (c) $15\sqrt{3}$ m (d) $10\sqrt{3}$ m

Ans. (a)

37. What is the angle between $\hat{i} + \hat{j} + \hat{k}$ and $\hat{j}$:

- (a) $\pi/6$ (b) $\pi/4$ (c) $\pi/3$ (d) None of these

Ans. (d)

38. A vector is represented by $3\hat{i} + \hat{j} + 2\hat{k}$. Its length in xy plane is :

- (a) 2 (b) $\sqrt{14}$ (c) $\sqrt{10}$ (d) $\sqrt{5}$

Ans. (c)

39. A force of 6N and another of 8N can be applied together to produce the effect of a single force of :

- (a) 1N (b) 11 N (c) 15 N (d) 20 N

Ans. (b)

40. If $\hat{n}$ is the unit vector of $(\vec{A} \times \vec{B})$ then

- (a) $\hat{n} = (\vec{A} \times \vec{B})$ (b) $\hat{n} = \vec{A}$

- (c) $\hat{n} = \hat{B}$ (d) $\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$

Ans. (d)